Module 4 – Introduction to DBMS

1. Introduction to SQL

- What is SQL, and why is it essential?

→ SQL is a standard language used to create, manage, and retrieve data from databases, and it is essential because it allows efficient and accurate data handling.

- Difference between DBMS and RDBMS

→ DBMS stores data in files without relationships, while RDBMS stores data in related tables using relational rules.

- Role of SQL in relational databases

→ SQL is used to create tables, insert data, modify records, and retrieve related data from relational databases.

- Key features of SQL

→ The key features of SQL include simplicity, security, portability, and support for multi-user access.

2. SQL Syntax

- What are the basic components of SQL syntax?

→ The basic components of SQL syntax include keywords, table names, column names, operators, and clauses.

- Write the general structure of an SQL SELECT statement.

→ The general structure of a SELECT statement is SELECT column_name FROM table_name.

- Explain the role of clauses in SQL statements.

→ Clauses are used to filter, group, sort, and limit the results of an SQL query.

3. SQL Constraints

- What are constraints in SQL?

→ Constraints are rules applied to table columns to ensure the accuracy and validity of data.

- How do PRIMARY KEY and FOREIGN KEY constraints differ?

→ A PRIMARY KEY uniquely identifies each record, while a FOREIGN KEY creates a relationship between two tables.

- What is the role of NOT NULL and UNIQUE constraints?

→ NOT NULL prevents empty values, and UNIQUE ensures that duplicate values are not allowed.

4. Main SQL Commands and Sub-commands (DDL)

- Define the SQL Data Definition Language (DDL).

→ DDL is a set of SQL commands used to define and modify the structure of database objects.

- Explain the CREATE command.

→ The CREATE command is used to create new databases, tables, and other database objects.

- What is the purpose of specifying data types and constraints?

→ Data types define the kind of data stored, and constraints ensure that only valid data is entered.

5. ALTER Command

- What is the use of the ALTER command in SQL?

→ The ALTER command is used to modify the structure of an existing table.

- How can you add, modify, and drop columns using ALTER?

→ Columns can be added, modified, or removed using the ADD, MODIFY, and DROP clauses.

6. DROP Command

- What is the function of the DROP command in SQL?

→ The DROP command is used to permanently delete a table or database from the system.

- What are the implications of dropping a table?

→ Dropping a table removes both its structure and all its data permanently.

7. Data Manipulation Language (DML)

- Define the INSERT, UPDATE, and DELETE commands.

→ INSERT adds new records, UPDATE modifies existing records, and DELETE removes records from a table.

- What is the importance of the WHERE clause?

→ The WHERE clause limits the affected rows and prevents unwanted changes to all records.

8. Data Query Language (DQL)

- What is the SELECT statement?

→ The SELECT statement is used to retrieve specific or all records from a database table.

- Explain the use of ORDER BY and WHERE clauses.

→ The WHERE clause filters records, and the ORDER BY clause sorts the result set.

9. Data Control Language (DCL)

- What is the purpose of GRANT and REVOKE?

→ GRANT gives permissions to users, and REVOKE removes previously given permissions.

- How do you manage privileges using these commands?

→ Privileges are managed by assigning or removing specific rights from users.

10. Transaction Control Language (TCL)

- What is the purpose of COMMIT and ROLLBACK?

→ COMMIT saves the transaction permanently, and ROLLBACK cancels the uncommitted changes.

- Explain how transactions are managed in SQL.

→ Transactions are managed to ensure data consistency and safe recovery from errors.

11. SQL Joins

- Explain the concept of JOIN in SQL.

→ A JOIN is used to combine related data from two or more tables.

- How are joins used to combine data?

→ Joins connect tables using common columns to retrieve related information.

12. SQL GROUP BY

- What is the GROUP BY clause?

→ The GROUP BY clause groups rows and applies aggregate functions to each group.

- Explain the difference between GROUP BY and ORDER BY.

→ GROUP BY groups records, while ORDER BY sorts records.

13. SQL Stored Procedures

- What is a stored procedure?

→ A stored procedure is a precompiled set of SQL statements stored in the database.

- Explain the advantages of stored procedures.

→ Stored procedures improve performance, security, and code reusability.

14. SQL Views

- What is a view in SQL?

→ A view is a virtual table created from a stored SQL query.

- Explain the advantages of using views.

→ Views improve security, simplify queries, and hide complex logic.

15. SQL Triggers

- What is a trigger in SQL?

→ A trigger is an automatic program that runs when a table event occurs.

- Explain the difference between INSERT, UPDATE, and DELETE triggers.

→ These triggers execute when a row is inserted, updated, or deleted.

16. Introduction to PL/SQL

- What is PL/SQL?

→ PL/SQL is a procedural extension of SQL that supports programming features.

- List the benefits of using PL/SQL.

→ PL/SQL provides better performance, error handling, and modular programming.

17. PL/SQL Control Structures

- What are control structures in PL/SQL?

→ Control structures are used to control the flow of execution in a program.

- How do control structures help in complex queries?

→ They allow decision making and repeated execution of statements.

18. SQL Cursors

- What is a cursor in PL/SQL?

→ A cursor is a pointer that allows row-by-row processing of query results.

- Difference between implicit and explicit cursors.

→ Implicit cursors are automatic, while explicit cursors are defined by the programmer.

19. Rollback and Commit Savepoint

- What is a SAVEPOINT?

→ A SAVEPOINT marks a point in a transaction to which a rollback can be done.

- When is it useful to use savepoints?

→ Savepoints are useful when only part of a transaction needs to be undone.